RAG Learning

C6 — Making Useful Chemicals

OCR GCSE Chemistry B (Twenty First Century Science, J258) — All papers (J258/01–04, Breadth and Depth in chemistry, Foundation and Higher) · (HT) = Higher tier only

This chapter is about making useful chemicals, from the laboratory bench to the chemical plant: salts from acids (C6.1), controlling how fast reactions go (C6.2), reversible reactions and yield (C6.3), and how fertilisers and other bulk chemicals are made on an industrial scale (C6.4).

C6.1 What useful products can be made from acids?

Acids are used to make cleaning products, medicines, food additives and, on a huge scale, fertilisers. Their reactions all produce **salts**. The salt's name comes from the acid: **hydrochloric** acid → **chlorides**, **nitric** acid → **nitrates**, **sulfuric** acid → **sulfates**.

Reactions of acids

General equation	Example	Observations
acid + metal → salt + hydrogen *(metals above hydrogen in the reactivity series, e.g. Mg, Zn, Fe; not Cu)*	$Mg + 2HCl \rightarrow MgCl_2 + H_2$ $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$	Fizzing; metal disappears; H_2 gives a squeaky pop with a lit splint
acid + carbonate → salt + water + carbon dioxide	$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2$	Fizzing; CO_2 turns limewater milky
acid + metal hydroxide (alkali) → salt + water	$HNO_3 + KOH \rightarrow KNO_3 + H_2O$	Temperature rises; no visible change without indicator

© RAG Learning · Not for redistribution

acid + metal oxide (base) → salt + water	$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$	Black solid dissolves; solution turns blue

Ionic equations: $Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$; $CO_3^{2-}(s \text{ or } aq) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$.

Formulae from ions

A compound has no overall charge, so combine ions so that the positive and negative charges balance.

Positive ions	Negative ions
+1: H^+, Na^+, K^+, Li^+, Ag^+, NH_4^+ (ammonium) +2: Mg^{2+}, Ca^{2+}, Ba^{2+}, Cu^{2+}, Zn^{2+}, Fe^{2+} +3: Al^{3+}, Fe^{3+}	−1: Cl^-, Br^-, I^-, OH^- (hydroxide), NO_3^- (nitrate) −2: O^{2-}, SO_4^{2-} (sulfate), CO_3^{2-} (carbonate) −3: PO_4^{3-} (phosphate)

Worked Example: Formula of calcium nitrate: Ca^{2+} needs two NO_3^- to balance → **$Ca(NO_3)_2$**. Ammonium sulfate: two NH_4^+ balance one SO_4^{2-} → **$(NH_4)_2SO_4$**. Aluminium sulfate: 2 × (+3) = 3 × (−2) → **$Al_2(SO_4)_3$**.

Making salts in the laboratory (PAG7)

A soluble salt from an insoluble base or carbonate (e.g. copper(II) sulfate crystals from copper(II) oxide and sulfuric acid):

1. Warm the dilute acid in a beaker (speeds up the reaction).
2. Add the insoluble solid a little at a time, stirring, until no more dissolves — the solid is **in excess**, so all the acid has reacted.
3. **Filter** to remove the excess solid; the filtrate is the salt solution.
4. **Evaporate** the filtrate gently in an evaporating basin until crystals start to form (point of crystallisation), then leave to cool so the salt **crystallises**.
5. Filter off the crystals, rinse with a little cold distilled water and **dry** them between filter papers or in a warm oven.

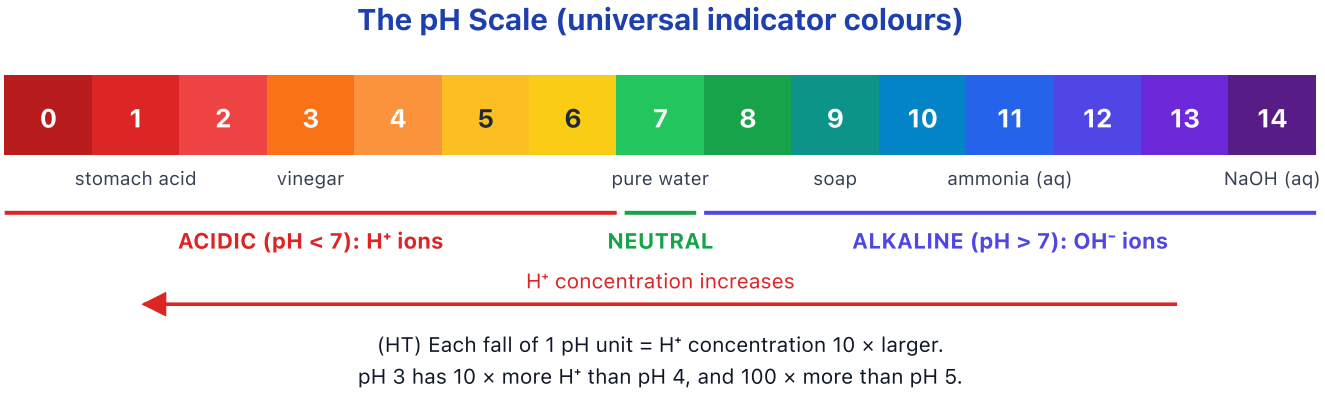
© RAG Learning · Not for redistribution

A soluble salt from an alkali (e.g. sodium chloride): both reactants are solutions, so excess cannot be filtered off. Do a titration with indicator to find the exact volumes, then repeat with the same volumes and *no* indicator, and crystallise.

An insoluble salt (e.g. barium sulfate): mix two solutions that contain its ions (precipitation), filter, wash the precipitate on the filter paper with distilled water and dry it.

Measuring pH

pH measures how acidic or alkaline a solution is. Use **universal indicator** (colour compared with a chart; quick and cheap but subjective and to whole numbers only) or a **pH meter** (probe; more precise, reads to decimals, records data, but needs calibrating).



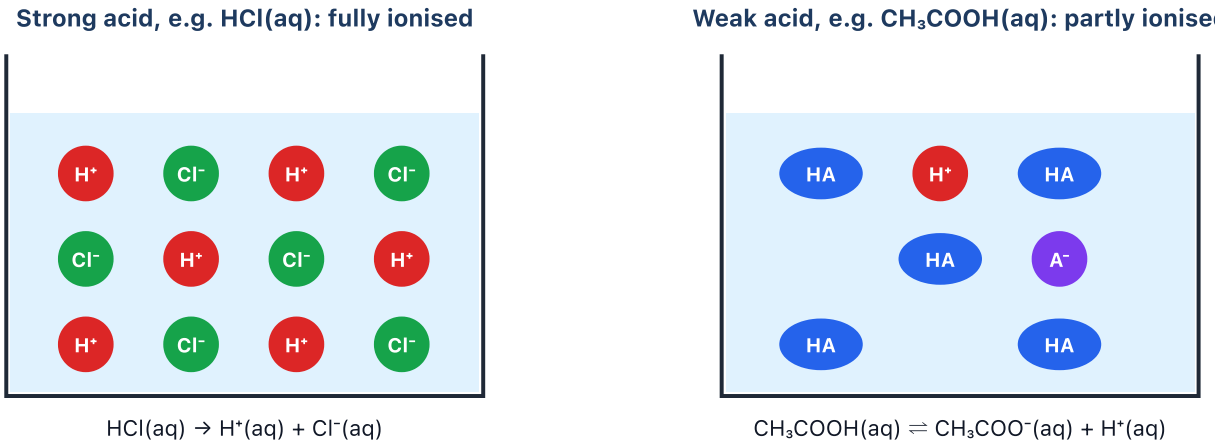
(HT) Each fall of 1 pH unit = H^+ concentration 10 × larger.
pH 3 has 10 × more H^+ than pH 4, and 100 × more than pH 5.

Strong and weak, dilute and concentrated acids (HT)

Strong / weak describes the **degree of ionisation**. A **strong acid** is **completely ionised** in water (e.g. hydrochloric, nitric, sulfuric). A **weak acid** is only **partially ionised** (e.g. ethanoic, citric, carbonic).
Dilute / concentrated describes the **amount of acid per unit volume** (mol/dm³). Both strong and weak acids can be made at any concentration.

© RAG Learning · Not for redistribution

Strong vs Weak Acid at the Same Concentration



HA = un-ionised CH_3COOH molecule; A^- = CH_3COO^- ion. Six acid units in each beaker: 6 H^+ (strong) vs 1 H^+ (weak).

- At the **same concentration**, a strong acid has a much higher H^+ concentration than a weak acid, so it has a **lower pH** (e.g. 0.1 mol/dm³ HCl is about pH 1; 0.1 mol/dm³ ethanoic acid is about pH 3).
- The strong acid is **more reactive**: it fizzes faster with magnesium or calcium carbonate, because there are more H^+ ions to collide with. With equal amounts of acid, both eventually make the **same total volume of gas** — the weak acid just takes longer, because more molecules ionise as H^+ ions are used up.
- "Dilute" does not mean "weak": a concentrated weak acid can have a lower pH than a very dilute strong acid.

pH and hydrogen ion concentration (HT)

- **Neutral** (pH 7): H^+ and OH^- concentrations are equal.
- **Acidic** (pH below 7): more H^+ than OH^-; the **higher the H^+ concentration, the lower the pH**.
- **Alkaline** (pH above 7): more OH^- than H^+; the lower the H^+ concentration, the higher the pH.
- When H^+ concentration increases by a **factor of 10**, the pH **decreases by 1**.

Worked Example: (HT) A solution of hydrochloric acid has pH 2. It is diluted until its H^+ concentration is 1/100 of the original. New pH? Each factor of 10 decrease raises pH by 1, so two factors of 10 → **pH 4**.

© RAG Learning · Not for redistribution

C6.2 How do chemists control the rate of reactions?

Controlling the rate lets chemists make a product as fast as is **safe** and economic. The rate can be changed by **temperature**, **concentration** (of solutions), **pressure** (of gases), **surface area** (of solids) and **catalysts**.

Collision theory

> Particles react only when they **collide** with at least a minimum amount of energy, the **activation energy**. The rate depends on the **frequency** of collisions and the **energy** of the collisions (the proportion that are successful).

Change	Effect on rate	Explanation (particle model)
Increase **temperature**	Increases	Particles move faster so collide more often; **more importantly**, more collisions have energy ≥ activation energy, so a larger proportion are successful
Increase **concentration** of a solution	Increases	More reacting particles in the same volume → more frequent collisions
Increase **pressure** of a gas	Increases	Same number of particles in a smaller volume → more frequent collisions
Increase **surface area** (smaller pieces, powder)	Increases	More particles exposed on the surface → more frequent collisions
Add a **catalyst**	Increases	Alternative pathway with a lower activation energy → more collisions are successful

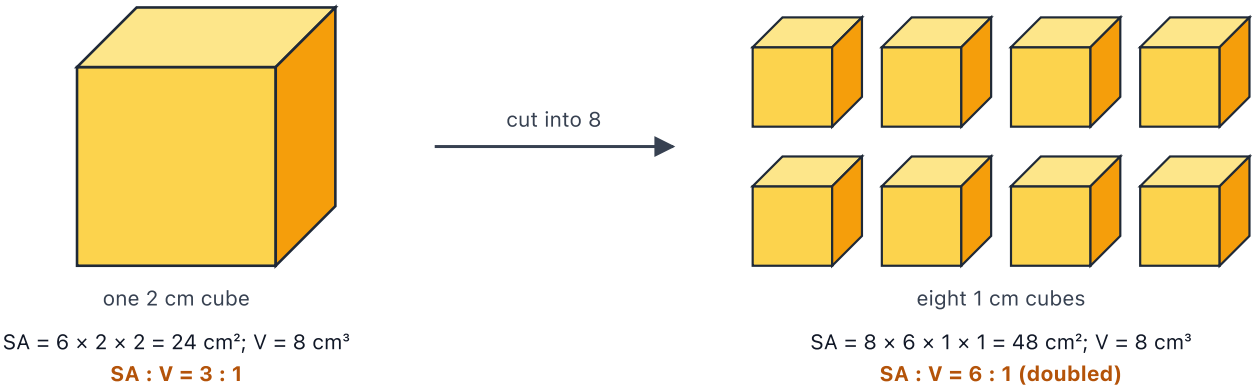
Ideas about Science (IaS3): the particle (collision) model is a scientific explanation used to account for all of these observed patterns.

Surface area to volume ratio

© RAG Learning · Not for redistribution

Cutting a solid into smaller pieces does not change its total volume but increases its total surface area, so the **surface area : volume ratio** goes up and more particles are available to collide.

Cutting Up a Solid Increases the Surface Area : Volume Ratio

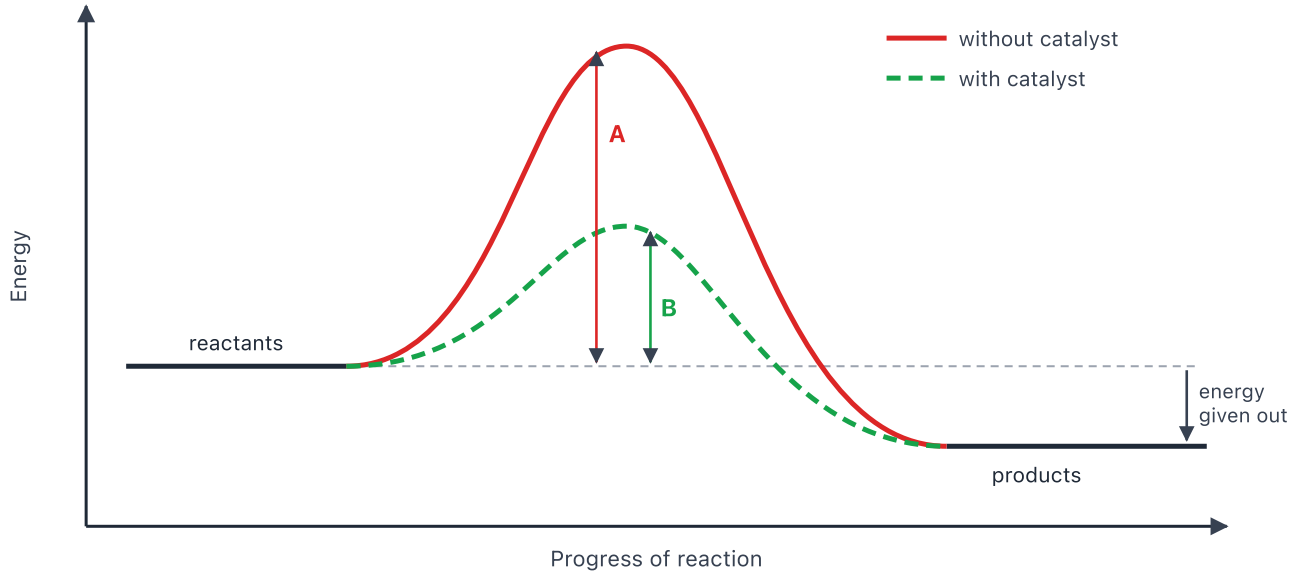


Catalysts

- A **catalyst** speeds up a reaction but is **not used up**: it is chemically unchanged, and its mass is the same at the end, so it can be recovered and reused.
- Only a **small amount** is needed, and catalysts are often **specific** to one reaction.
- It works by providing an **alternative reaction pathway with a lower activation energy**, so a greater proportion of collisions are successful.
- **Identifying a catalyst:** it is not included in the overall equation (often written over the arrow), and it is present unchanged at the end. E.g. manganese(IV) oxide in $2H_2O_2 \rightarrow 2H_2O + O_2$; iron in the Haber process; enzymes in living things.
- **Ideas about Science (IaS4):** catalysts let industrial processes run at lower temperatures, saving energy and raw materials and cutting emissions — more sustainable, "green" chemistry.

© RAG Learning · Not for redistribution

Reaction Profile: Effect of a Catalyst (exothermic reaction)



A = activation energy without catalyst; B = activation energy with catalyst (lower).
The catalyst does not change the energy of the reactants or products, so the energy given out is the same.

Measuring the rate of a reaction (PAG8)

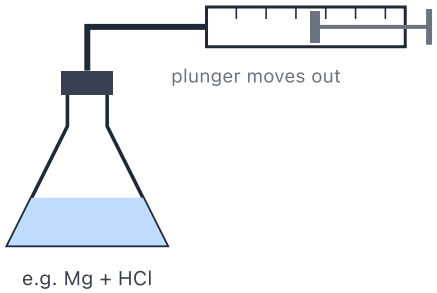
Rate can be found by measuring how fast a **product forms** or a **reactant is used up**.

- **Gas produced — volume:** collect the gas in a **gas syringe**, or **over water** in an upturned measuring cylinder or burette, and record the volume at regular time intervals.
- **Gas produced — mass:** stand the flask on a **balance** and record the loss in mass as gas escapes (a cotton wool plug lets gas out but stops acid spray). Works poorly for H_2 because it is so light.
- **Colour change (HT):** time how long a colour takes to appear or disappear, by eye or with a **colorimeter** that measures how much light passes through the solution.
- **Formation of a precipitate (HT):** time how long it takes for a cross under the flask to disappear as the solution goes cloudy, e.g. sodium thiosulfate + hydrochloric acid → a yellow precipitate of sulfur. A light sensor/colorimeter removes the judgement of the eye.

© RAG Learning · Not for redistribution

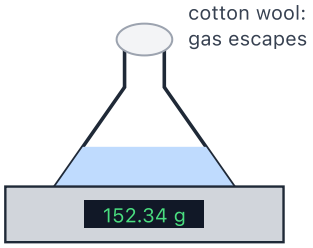
Three Ways to Follow a Reaction

Gas syringe



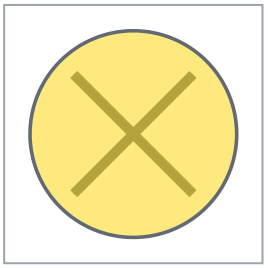
record volume of gas every 10 s

Mass loss on a balance



record mass every 10 s; mass lost = mass of gas

Disappearing cross (HT)

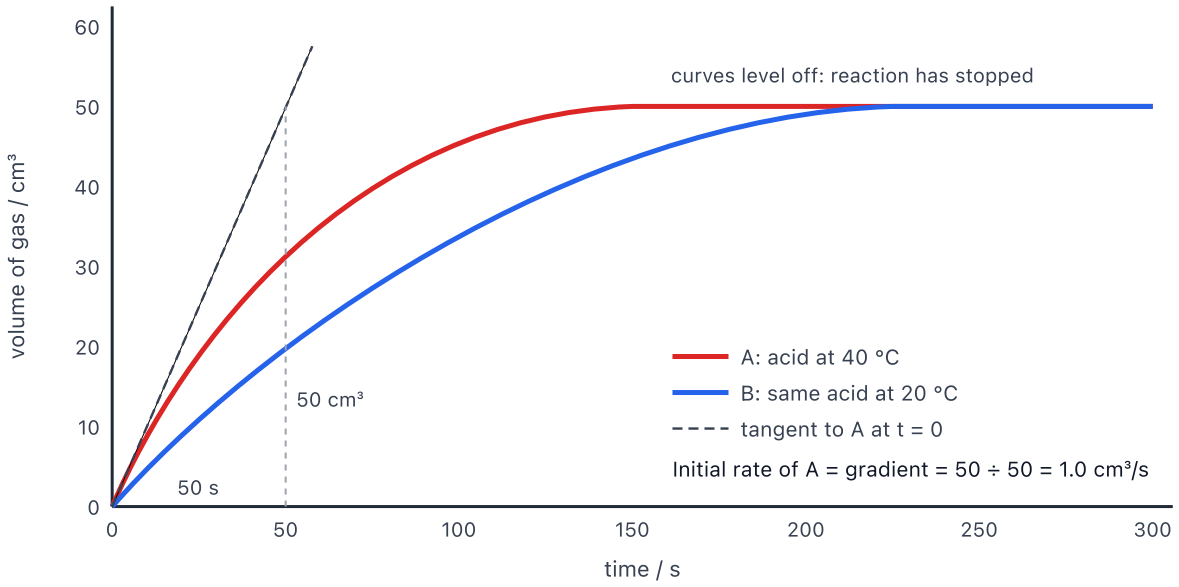


flask on a paper cross, viewed from above

time until the cross can no longer be seen (sulfur ppt)

Interpreting rate graphs

Volume of Gas against Time (illustrative data)



- The curve is **steepest at the start** (highest concentration of reactants, fastest rate) and gets less steep as reactants are used up.
- When the curve becomes **horizontal**, the reaction has stopped because the limiting reactant is used up.
- A **steeper** curve means a faster rate. If the amounts of reactants are the same, the curves finish at the **same final volume**.
- The **rate at any moment** = the **gradient of the tangent** to the curve at that time (change in y ÷ change in x).

© RAG Learning · Not for redistribution

- **Mean rate** = amount of product formed (or reactant used) ÷ time taken. From the graph, curve A makes 50 cm³ in about 150 s: mean rate = 50 ÷ 150 = 0.33 cm³/s.

mean rate of reaction = quantity of product formed ÷ time taken

units: cm³/s or g/s rate ∝ 1 ÷ time (when timing a fixed change, e.g. a disappearing cross)

Graphs of reaction conditions against rate (HT)

Plot a measure of rate (e.g. 1/time, or initial gradient) against the condition you changed.

- **Rate vs concentration** is often a straight line through the origin: doubling the concentration doubles the frequency of collisions, so roughly doubles the rate (rate is proportional to concentration).
- **Rate vs temperature** curves upwards: the rate increases more and more steeply as temperature rises (a common rule of thumb is that it roughly doubles for every 10 °C rise).
- **Rate vs surface area** increases as particle size decreases.

Worked Example: Using proportionality (M1c): in a disappearing-cross experiment the cross vanishes in 80 s at a thiosulfate concentration of 0.10 mol/dm³ (illustrative data). Predict the time at 0.20 mol/dm³.
Doubling concentration doubles the rate; rate ∝ 1/time, so time halves: **40 s**. Relative rate = 1/40 = 0.025 s^{-1} compared with 1/80 = 0.0125 s^{-1}.

Practical work (PAG8): investigate the effect of surface area, concentration and temperature on rate (e.g. marble chips of different sizes with acid, measuring gas volume or mass loss), and compare two methods of following the same reaction.

Enzymes

- **Enzymes** are **proteins** that act as **biological catalysts** in living organisms (link to B3.1).
- They work best within a narrow range of **temperature** and **pH** (the optimum); outside it they work slowly or are **denatured** (their shape changes permanently).
- Industrial uses: yeast enzymes in **fermentation** to make ethanol; enzymes in **biological washing powders** that work at low temperatures; making sugar syrups for food from starch.

© RAG Learning · Not for redistribution

- Enzymes **limit** the conditions that can be used, but this is often an **advantage**: processes run at low temperatures and normal pressure, so energy demand and costs are lower.

C6.3 What factors affect the yield of chemical reactions?

Reversible reactions

Some reactions can go in both directions, depending on the conditions. They are shown with the symbol ⇌.

- Heating ammonium chloride: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$ — it decomposes when heated, and the gases recombine on cooling.
- Hydrated copper(II) sulfate (blue) ⇌ anhydrous copper(II) sulfate (white) + water. Heating drives water off; adding water turns it blue again (and releases heat).
- The Haber process: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.

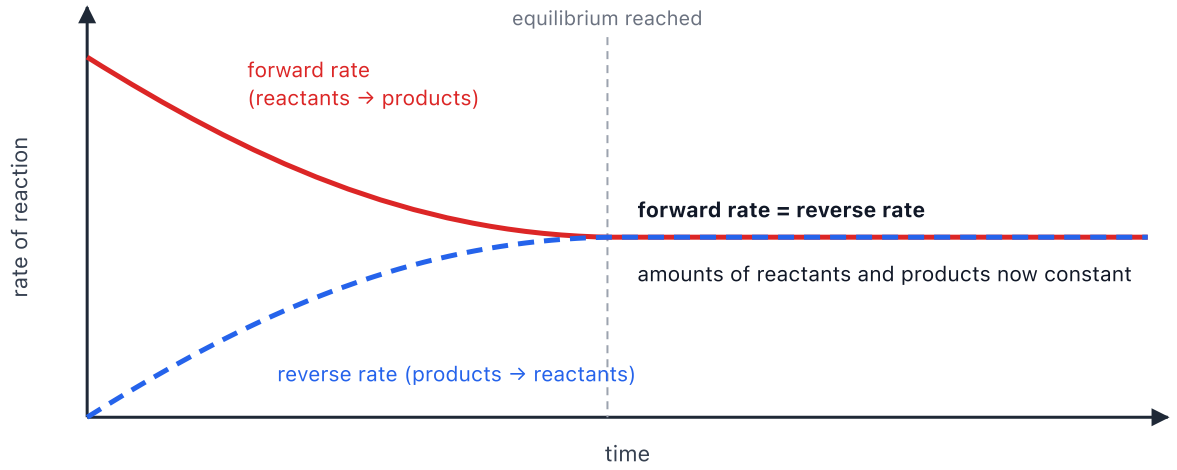
In a closed system a reversible reaction **never reaches 100% yield**, because as products form some of them turn back into reactants. In industry this wastes reactants and means extra stages are needed to separate products from unreacted reactants.

Dynamic equilibrium

> In a closed system, a reversible reaction reaches **dynamic equilibrium** when the **rate of the forward reaction equals the rate of the reverse reaction**. Both reactions are still happening, so the amounts (concentrations) of reactants and products stay **constant** — but they are not necessarily equal.

© RAG Learning · Not for redistribution

Reaching Dynamic Equilibrium (closed system)



Forward rate falls as reactants are used up; reverse rate rises as products build up.

Changing the position of equilibrium (HT)

If the conditions of a system at equilibrium are changed, the position of equilibrium shifts to **oppose the change**:

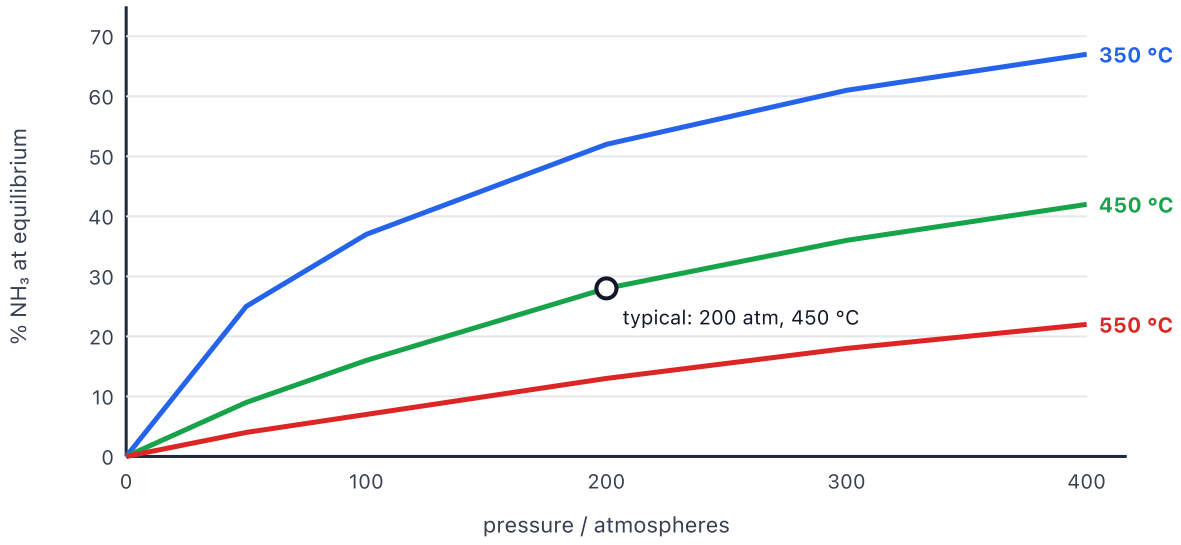
Change	Equilibrium shifts...	Haber process $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (forward reaction exothermic)
Increase concentration of a reactant	Towards the products (to use it up)	Adding N_2 or H_2, or removing NH_3, increases the yield
Increase temperature	In the **endothermic** direction	Shifts left → **lower** yield of NH_3 (but faster rate)
Decrease temperature	In the **exothermic** direction	Shifts right → higher yield (but slower rate)
Increase pressure (gases)	Towards the side with **fewer gas molecules**	4 molecules → 2 molecules, so shifts right → **higher** yield
Add a catalyst	**No change** in position	Equilibrium reached faster; yield unchanged

- **Catalysts** speed up the forward and reverse reactions equally, so they increase the rate but **do not affect the yield**.

© RAG Learning · Not for redistribution

- **Disadvantages of very high temperatures or pressures:** large energy costs to heat or compress the gases; expensive, thick-walled equipment that can still fail under extreme conditions; greater safety risk (explosion, leaks). High temperature may also lower the yield of an exothermic reaction.
- Chemical engineers therefore choose conditions that give the best **economic outcome**: a good enough yield, fast enough, safely, with acceptable energy and equipment costs and as little waste as possible.

Equilibrium Yield of Ammonia (approximate, illustrative)



Higher pressure → higher yield; higher temperature → lower yield.

Ideas about Science (IaS2): make predictions from data and graphs like this about the yield of a product. **(IaS4):** weigh up the risks and costs of different operating conditions in an ammonia plant.

C6.4 How are chemicals made on an industrial scale?

Fertilisers and the importance of N, P and K

Plants need **nitrogen (N)**, **phosphorus (P)** and **potassium (K)** compounds for healthy growth (N for proteins and leaf growth, P for roots and DNA, K for flowers, fruit and enzymes). Crops take these elements from the soil and they are removed when the crop is harvested, so **fertilisers** are added to replace them. NPK fertilisers are formulations containing salts such as ammonium nitrate

© RAG Learning · Not for redistribution

NH_4NO_3, ammonium sulfate $(NH_4)_2SO_4$, ammonium phosphate $(NH_4)_3PO_4$ and potassium chloride KCl.

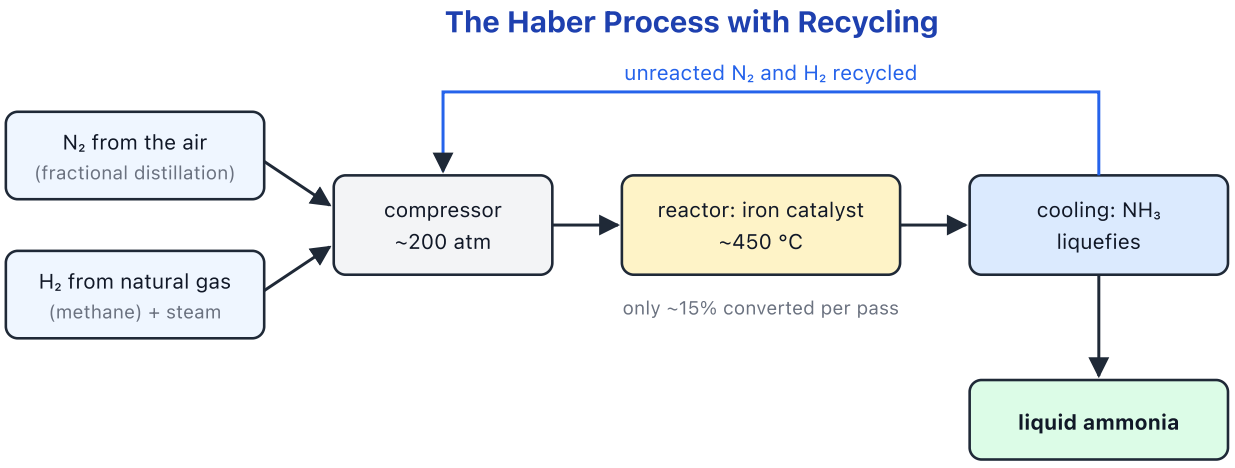
Benefits and costs of fertilisers

- **Demand and supply of food:** the world's population could not be fed without synthetic fertilisers — they greatly increase crop yields from the same land. (IaS4: fertiliser production enhances the quality of people's lives.)
- **Synthetic vs natural fertilisers:** natural fertilisers (manure, compost) recycle nutrients and add organic matter, but are not available in large enough amounts, are bulky and hard to transport and manage, and their composition varies. Synthetic fertilisers have a known, controlled composition and are concentrated, but making them uses non-renewable natural gas, lots of energy, and releases CO_2.
- **Over-use and eutrophication:** excess fertiliser washes into rivers and lakes. Algae and water plants grow rapidly, block light, die and are decomposed by bacteria that use up the dissolved oxygen — fish and other organisms die.

The Haber process

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \text{forward reaction exothermic}$$

Ammonia is the starting point for almost all nitrogen fertilisers, so the Haber process is vital to agriculture.



A continuous process: ammonia is removed as a liquid; unreacted gases go round again so little is wasted.

© RAG Learning · Not for redistribution

Choosing the conditions for the Haber process (HT)

- **Raw materials and feedstocks:** nitrogen from the **air** (by fractional distillation of liquid air); hydrogen from **natural gas (methane) and steam**: $CH_4 + H_2O \rightarrow CO + 3H_2$. Air is free and abundant; natural gas is non-renewable and its price affects the cost of ammonia.
- **Temperature (~450 °C):** the forward reaction is exothermic, so a low temperature gives a higher equilibrium yield — but the rate would be far too slow. 450 °C is a **compromise** between a reasonable yield and a fast enough rate.
- **Pressure (~200 atm):** higher pressure increases both the yield (fewer gas molecules on the right) and the rate. But very high pressures need expensive, thick-walled plant, cost a lot of energy to compress the gases and raise safety risks, so a compromise pressure is used.
- **Catalyst (iron):** increases the rate so a useful amount is made quickly at a moderate temperature; it does not change the yield.
- **Separation and recycling:** the reaction mixture is cooled, ammonia condenses to a liquid and is removed, and unreacted N_2 and H_2 are **recycled**. In practice the gases never reach equilibrium in the reactor (only ~15% conversion per pass), but recycling gives an overall conversion of about 98%.

> **Rate vs equilibrium trade-off (HT):** for an exothermic reaction, a lower temperature gives a better equilibrium yield but a slower rate. Industry chooses conditions that make the most product **per hour** at an acceptable cost — a fast rate at a moderate yield, with recycling, often beats a high yield made slowly. The same trade-off applies to other important processes, e.g. the Contact process for sulfuric acid ($2SO_2 + O_2 \rightleftharpoons 2SO_3$, also exothermic).

Atom economy

> **Atom economy** is the percentage of the mass of all the atoms in the reactants that ends up in the **desired product**. It is a theoretical value calculated from the balanced equation, used alongside yield and rate when processes are evaluated.

atom economy = (mass of atoms in desired product ÷ total mass of atoms in reactants) × 100

using the relative formula masses (× the numbers in the balanced equation)

© RAG Learning · Not for redistribution

Worked Example: Making ammonium sulfate: $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$. Only one product, so atom economy = **100%**.
Making calcium oxide (lime): $CaCO_3 \rightarrow CaO + CO_2$. Atom economy = 56 ÷ 100 × 100 = **56%**; the CO_2 is a waste product.

A high atom economy means less waste and a more sustainable process (fewer raw materials used per kg of product).

Choosing a reaction pathway (HT)

Given data, explain why a particular route is chosen and evaluate its **sustainability** — can it meet demand now without long-term harm to the environment? Consider: **atom economy**, **yield**, **rate**, **equilibrium position**, **usefulness of by-products**, whether raw materials are **renewable**, **competing uses** for those raw materials, the nature and amount of **waste**, and the **energy** needed.

Making ethanol	Fermentation of glucose	Hydration of ethene
Equation	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$	$C_2H_4 + H_2O \rightarrow C_2H_5OH$
Atom economy	92 ÷ 180 × 100 = 51%	100%
Raw materials	Renewable (sugar crops) — but competes with land for food	Non-renewable (crude oil)
Conditions / rate	~35 °C, normal pressure, enzymes in yeast; slow, batch	~300 °C, 60–70 atm, catalyst; fast, continuous
By-products / purity	CO_2 (could be captured and sold); dilute ethanol needs distilling	No by-product; unreacted ethene recycled; purer product

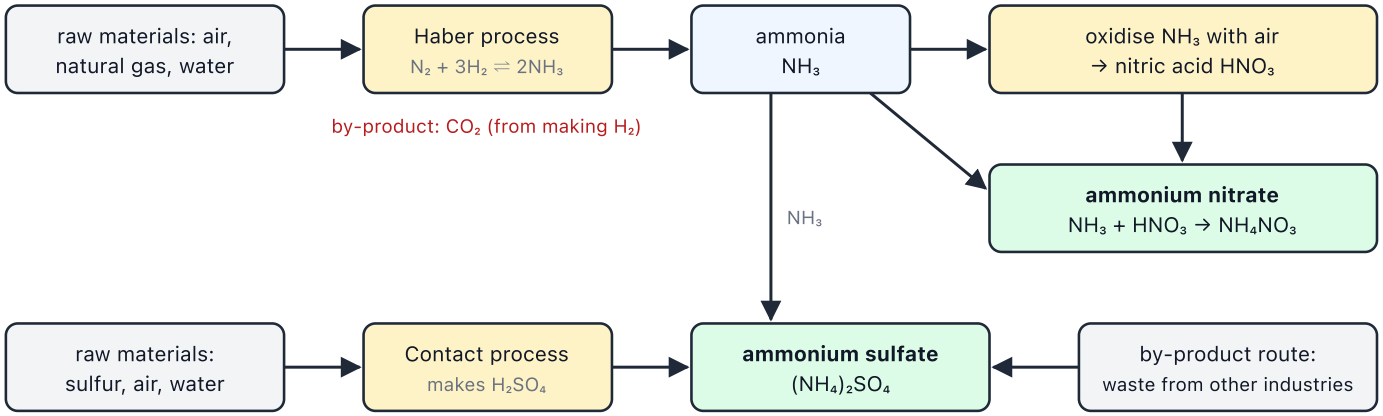
Industrial production of fertilisers

Fertiliser manufacture is a set of **integrated processes** using several raw materials (air, natural gas, water, sulfur, phosphate rock). **Process flow charts** summarise the raw materials, stages,

© RAG Learning · Not for redistribution

products, by-products and waste.

Process Flow Chart: Making Nitrogen Fertilisers



Industrial plants run continuously; demand for ammonium sulfate is met by more than one route.

- **More than one process meets demand:** ammonium sulfate can be made by reacting ammonia with sulfuric acid (as in the lab), or obtained as a **by-product or waste product** of other processes (e.g. nylon manufacture, coke ovens, cleaning sulfur dioxide from power-station flue gases). Finding uses for by-products makes industry more sustainable.
- **Batch vs continuous:** lab processes make chemicals in **batches**; industrial processes are usually **continuous** — raw materials fed in and product removed all the time, 24 hours a day.

Stage	Laboratory preparation of ammonium sulfate	Industrial production
Reactants	Bought ammonia solution and dilute sulfuric acid	Ammonia from the Haber process, sulfuric acid from the Contact process (or a by-product source)
Synthesis	Titration in a flask, using indicator to find the right volumes	Ammonia gas fed continuously into concentrated sulfuric acid in a large reactor
Monitoring	Indicator / pH paper	Automatic sensors and computer control
Separation	Evaporate, crystallise, filter, dry by hand	Continuous crystallisers, centrifuges and dryers

© RAG Learning · Not for redistribution

Scale, energy, waste	Grams; small energy use; waste poured away safely	Thousands of tonnes; heat from the exothermic reaction recovered; waste treated or sold
Purity testing	Melting point, chemical tests	Automated quality control sampling

Ideas about Science (IaS3): industrial processes are first modelled in the laboratory and with computer simulations before decisions are made about full-scale production. **(IaS4):** evaluate industrial processes in terms of sustainability, risk, costs and benefits. **Practical:** the laboratory preparation and purification of fertiliser salts (PAG7).

> **Exam Tip:** For "explain the conditions" questions on the Haber process, always give both sides: what the condition does to the **yield** and what it does to the **rate** (and cost/safety), then say why the chosen value is a compromise.

C6 Summary & Key Equations

- acid + metal → salt + hydrogen; acid + carbonate → salt + water + CO_2; acid + base/alkali → salt + water
- (HT) strong = fully ionised; weak = partly ionised; concentrated/dilute = amount per volume
- (HT) H^+ concentration ×10 → pH falls by 1
- Rate increases with temperature, concentration, pressure, surface area and a catalyst
- mean rate = quantity of product ÷ time; instantaneous rate = gradient of the tangent
- Catalysts lower the activation energy; they are not used up and do not change the yield
- Dynamic equilibrium: forward rate = reverse rate in a closed system
- Haber: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, iron catalyst, ~450 °C, ~200 atm, recycle unreacted gases
- atom economy = mass of atoms in desired product ÷ total mass of atoms in reactants × 100

> **Exam Tip:** Don't confuse *yield* with *rate*. A catalyst affects only the rate; temperature and pressure can change both. And never write that a catalyst "takes part and is used up".

© RAG Learning · Not for redistribution

Spec Checklist — C6

Spec	You should be able to...
C6.1.1	Recall that acids react with some metals and with carbonates, and write equations predicting products
C6.1.2	Describe practical procedures to make salts using filtration, evaporation, crystallisation and drying (PAG7)
C6.1.3	Use the formulae of common ions to deduce the formula of a compound
C6.1.4	Recall that relative acidity and alkalinity are measured by pH, using universal indicator and pH meters
C6.1.5 (HT)	Use and explain dilute/concentrated (amount of substance) and weak/strong (degree of ionisation), including differences in reactivity with metals and carbonates
C6.1.6 (HT)	Use the idea that as H^+ concentration increases by a factor of ten, pH decreases by one
C6.1.7 (HT)	Describe neutrality and relative acidity/alkalinity in terms of H^+ concentration and pH (whole numbers)
C6.2.1	Describe the effect on rate of temperature, concentration, pressure and surface area
C6.2.2	Explain the effects of temperature, concentration and pressure in terms of frequency and energy of collisions
C6.2.3	Explain the effect of the size of pieces of a solid in terms of surface area to volume ratio
C6.2.4	Describe the characteristics of catalysts and their effect on rate
C6.2.5	Identify catalysts in reactions
C6.2.6	Explain catalytic action in terms of activation energy

© RAG Learning · Not for redistribution

C6.2.7	Suggest practical methods for determining rate: gas syringe or collection over water; mass change on a balance; (HT) colour change; (HT) formation of a precipitate (PAG8)
C6.2.8	Interpret rate of reaction graphs
C6.2.9 (HT)	Interpret graphs of reaction conditions versus rate
C6.2.10	Use arithmetic computation and ratios when measuring rates
C6.2.11	Draw and interpret graphs from data to determine rate
C6.2.12	Determine gradients of graphs as a measure of rate of change
C6.2.13	Use proportionality when comparing factors affecting rate
C6.2.14	Describe the use of enzymes as catalysts in biological systems and some industrial processes
C6.3.1	Recall that some reactions can be reversed: shown by ⇌; do not reach 100% yield in closed systems
C6.3.2	Recall that dynamic equilibrium occurs when forward and reverse rates are equal
C6.3.3 (HT)	Predict the effect of concentration, temperature and pressure on equilibrium position and suggest conditions; catalysts increase rate not yield; disadvantages of very high temperatures or pressures
C6.4.1	Recall the importance of N, P and K compounds in agriculture
C6.4.2	Explain the importance of the Haber process and the benefits and costs of fertilisers: food supply and demand; synthetic vs natural fertilisers; eutrophication
C6.4.3 (HT)	Explain the commercial Haber conditions: raw materials (N_2 from air, H_2 from natural gas and steam); effect of catalyst, temperature and pressure on yield and rate; separation and recycling

© RAG Learning · Not for redistribution

C6.4.4 (HT)	Explain the trade-off between rate of production and position of equilibrium in industrial processes
C6.4.5	Define the atom economy of a reaction
C6.4.6	Calculate atom economy from a balanced equation
C6.4.7	Use arithmetic computation when calculating atom economy
C6.4.8 (HT)	Explain why a reaction pathway is chosen using atom economy, yield, rate, equilibrium position and by-products, and evaluate sustainability
C6.4.9	Describe industrial fertiliser production as integrated processes: more than one route; by-product sources; process flow charts; batch vs continuous
C6.4.10	Compare industrial production of fertilisers with laboratory syntheses of the same products

© RAG Learning · Not for redistribution